



CSCE 411/811

Introduction to Data Modeling

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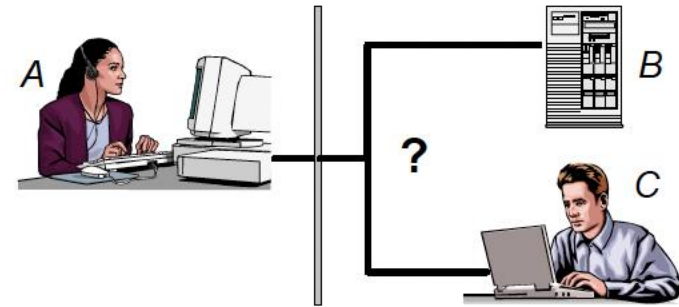
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College of Engineering

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Information Levels



- (external: end users, modelers, domain experts)
- Conceptual model: designed for clear communication between external roles.
- Logical model: abstract data structure and operations supporting conceptual model
- Physical model: specifics in DBMS, eg. Indexing, data types in relational tables
- Logical and physical models belong to internal levels for information system.
- Conceptual model can be stable during system updates.

Conceptual modeling

- require communication and help communication
- closest to natural languages
- (1)fact types/entities/objects
 name and attributes
- (2)constraints/relationships
 population A; population A to population B
- (3)derivation rules / semantic constraints
 something in population follows some rule

Student enrollment system

- Entities:

Student

Course

Enrollment

- Attributes:

Student: Student_ID, First_Name, Last_Name, Email, Phone_Number

Course: Course_ID, Course_Name, Instructor, Credited_hours

Enrollment: Enrollment_ID, Student_ID, Course_ID, Enrollment_Date

- Relationships:

Each student can enroll in several courses.

Each courses are offered to one student at one specific Enrollment_date.

Enrollment represents the association between Student and Course.

Student enrollment system

- Think more about semantic rules.
- For example, do you want to set up a capacity for a course?
- Then you need to add the following information:
- Derivation/Semantic rules:
The number of students enrolled in class cannot be larger than course capacity.
- Course: Course_ID, Course_Name, Instructor, Credited_hours, Capacity

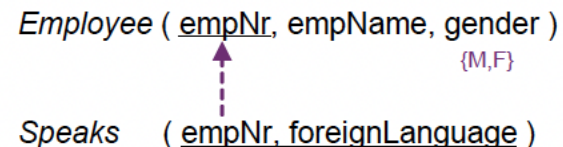
Conceptual Information Processor (CIP)

- CIP: consider the operation such as: add, insert, delete, update, search... from a conceptual way.
- Example:
- Add enrollment record for a (student, course) pair:
$$\text{total_student_count} = \text{count}(\text{student}, \text{course}) + 1$$

insert this enrollment record when
total_student_count hasn't reach course capacity.

Logical Modeling

- Generic logical data model (eg. Relational, networks, hierarchic...)
- Diagrams are useful tools to show the logical model
- More abstract than natural languages
- Test using data records
- Decide dependencies and keys



Student enrollment system

- Entities with Attributes:

Student: Student_ID (PK), First_Name, Last_Name, Email, Phone_Number

Course: Course_ID (PK), Course_Name, Instructor, Credits, Capacity

Enrollment: Enrollment_ID (PK), Student_ID (FK), Course_ID (FK), Enrollment_Date

- Relationships:

Many-to-Many relationship between Student and Course through Enrollment.

Student enrollment system

Q

Can you generate a text based ER diagram according to the following information: ""Entities with Attributes:

Student: Student_ID (PK), First_Name, Last_Name, Email, Phone_Number

Course: Course_ID (PK), Course_Name, Instructor, Credits, Capacity

Enrollment: Enrollment_ID (PK), Student_ID (FK), Course_ID (FK), Enrollment_Date

Relationships:

Many-to-Many relationship between Student and Course through Enrollment.""

Student enrollment system

- Looks a little bit weird. You can regenerate it with prompt easily.

```
+-----+           +-----+
|  Student  |       | Course |
+-----+           +-----+
| Student_ID |<-----| Course_ID |
| First_Name |       | Course_Name |
| Last_Name  |       | Instructor  |
| Email      |       | Credits     |
| Phone_Number|      | Capacity   |
+-----+           +-----+
|           |       |           |
|           |       |           |
|           |       |           |
|           |       |           |
+-----+           +-----+
| Enrollment |      | Enrollment |
+-----+           +-----+
| Enrollment_ID|    | Student_ID |
| Student_ID |----->| Course_ID |
| Course_ID  |       | Enrollment_Date|
+-----+           +-----+
```

Student enrollment system

Q

Can you revise the above diagram? We only need one "Enrollment" table, and the StudentID from "Enrollment" is from "Student" table, the CourseID from "Enrollment" is from "Course" table, there is no relationship between student and course tables.



Certainly! Here's the revised text-based representation of the ER diagram based on the updated information:

lua

 Copy code

```
+-----+      +-----+
|  Student  |    |  Course  |
+-----+      +-----+
| Student_ID |    | Course_ID |
| First_Name |    | Course_Name |
| Last_Name  |    | Instructor |
| Email      |    | Credits    |
| Phone_Number|    | Capacity  |
+-----+      +-----+
|           |    |           |
|           |    |           |
|           |    |           |
|           |    |           |
+-----+      +-----+
| Enrollment |    |           |
+-----+      +-----+
| Enrollment_ID|    |           |
| Student_ID  |----->|           |
| Course_ID   |    |           |
+-----+      +-----+
```

Physical Modeling

- DBMS level design
- Put indexing, or sorting features
- Define data representation/storage types
- Normalize the table(s) if it wasn't designed well
- Implement keys and foreign keys
- Triggers and transactions if there is a need
- Physical models are very technical. Physical processes are SQL codes.

Student enrollment system

- Indexes:

Index on Student_ID in Student table

Index on Course_ID in Course table

Index on Student_ID and Course_ID in Enrollment table

- Constraints:

Unique constraint on Student_ID in Student table

Unique constraint on Course_ID in Course table

Unique constraint on Enrollment_ID in Enrollment table

- Foreign key constraints on Student_ID and Course_ID in Enrollment table

- Normalization:

Ensure data is normalized to minimize redundancy and maintain data integrity.

Student enrollment system

- Print table structures (multiple styles)

Q

Please display the table structures for the above tables.



Certainly! Here are the table structures for the student, course, and enrollment tables:

Table: Student

sql Copy code

Column	Data Type	Constraints
Student_ID	INT	Primary Key
First_Name	VARCHAR	
Last_Name	VARCHAR	
Email	VARCHAR	
Phone_Number	VARCHAR	

Table: Student

Column	Data Type	Constraints
Student_ID	INT	Primary Key
First_Name	VARCHAR	
Last_Name	VARCHAR	
Email	VARCHAR	
Phone_Number	VARCHAR	

Table: Course

Column	Data Type	Constraints
Course_ID	INT	Primary Key
Course_Name	VARCHAR	
Instructor	VARCHAR	
Credits	INT	
Capacity	INT	

Table: Enrollment

Column	Data Type	Constraints
Enrollment_ID	INT	Primary Key
Student_ID	INT	Foreign Key
Course_ID	INT	Foreign Key
Enrollment_Date	DATE	

Data Modeling Work Cycle

- Feasibility study
- External requirements analysis
- Conceptual design: data, processes
- Logical design: data, processes
- Basic physical design: data, processes
- Basic external design: data; processes
- Prototyping
- Completion of design
- Implementation of production version
- Testing and validation
- Release of software, documentation, and training
- Maintenance

